MILITARY SPECIFICATION

# TEST SET, ARMAMENT WIRING SYSTEM, SPECIAL WEAPON AN/GWM-4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

* 1.1 This specification covers one type of electrical ground support equipment designated as the Test Set, Armament Wiring System, Special Weapon AN/GWM-4, hereinafter referred to as the test set.

## 2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-E-4158 | Electronic Equipment, Ground; General Requirements For |
| MIL-D-1000 | Drawings, Engineering And Associated Lists |
| MIL-E-17555 | Electronic And Electrical Equipment, Accessories, And Repair Parts: Packaging And Packing Of |
| MIL-C-25200 | Cable Assemblies, Special Weapons, Electrical, General Requirements For |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-130 | Identification Marking Of US Military Property |
| MIL-STD-810 | Environmental Test Methods For Aerospace And Ground Equipment |
| MIL-STD-831 | Test Reports, Preparation Of |

DRAWINGS

Guardian Electric Manufacturing Company

| | |
|---|---|
| No. 111-100 | Test Set, Armament Wiring System, Special Weapon AN/GWM-4 |

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute, Inc.

ANSI Y32.16 Reference Designations for Electrical and Electronics Parts and Equipments

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York NY 10018.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provisions for preproduction testing.

3.2 General. The requirements of MIL-E-4158 and MIL-C-25200 are applicable as requirements of this specification. When conflict occurs with this specification, the requirements of this specification shall govern.

3.3 Components. The Test Set, Armament Wiring System, Special Weapon AN/GWM-4 shall consist of the following:

a. Analyzer, Test Set TS-1304/GWM-4 (Basic Test Set)

b. Cover, Test Set CW-520/GWM-4

c. Cable Assembly, Special Purpose, Electrical, Branched CX-4828/U

d. Cable Assembly, Special Purpose, Electrical, Branched CX-4829/U

e. Cable Assembly, Special Purpose, Electrical, CX-8932/U

* 3.3.1 The Cable Assembly, Special Purpose, Electrical, Branched CX-4828/U shall be referred to hereafter as the fighter cable.

3.3.2 The Cable Assembly, Special Purpose, Electrical, Branched CX-4829/U shall be referred to hereafter as the bomber cable.

3.3.3 The Cover, Test Set CW-520/GWM-4 shall provide storage and carrying space for the cables.

* 3.3.4 The Cable Assembly, Special Purpose, Electrical CX-8932/U shall be used to adapt Cable Assembly, Special Purpose, Electrical, Branched CX-4828/U to aircraft modified for PAL capability.

3.4 Design and construction. The design and construction of the test set shall be in conformance with Guardian Electric Manufacturing Company drawing No. 111-100 and subsidiary drawings. Bomber and fighter cables shall be in accordance with MIL-C-25200.



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* 3.4.1 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

* 3.5 Performance. The test set shall be capable of performance of the necessary tests and checks to assure proper functioning of the applicable bomber and fighter wiring systems.

* 3.6 Environmental requirements. The test set shall be capable of satisfactory performance during or after subjection to the following environmental conditions as specified:

a. High temperature, up to 65° Centigrade (C) (149° Fahrenheit (F)) for a period of 50 hours.

b. Low temperature, between -54°C (-65°F) and -62°C (-80°F) for a period of 72 hours.

c. Humidity of 95 percent with temperature ranging from 20°C (68°F) to 65°C (149°F).

d. Rain, as described in 4.6.

e. Sand and dust as encountered under service conditions in desert areas.

f. Salt fog comparable to salt sea atmosphere.

g. Altitude up to 6,000 feet above sea level.

h. Vibration to be encountered during actual service and transportation.

i. Shock equal to being dropped from a vehicle when in transport configuration.

j. Explosive mixtures in surrounding atmosphere.

k. Fungus as found in tropical or damp climates.

3.7 Identification of product. Equipment assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.8 Reference designations. Reference designations used for markings within the unit shall be in accordance with ANSI Y32.16. Markings shall not adversely affect the leakage path between conductors, or any other factor of performance. Reference symbols shall be employed to identify each part for its particular circuit application. The same reference symbol shall be used for identifying a particular part on all types of data, including drawings, diagrams, and spare parts lists prepared for the equipment. When practicable, markings shall be placed on the supporting structure adjacent to each replaceable part in such a position as to be readily viewed.

3.9 Workmanship. The test set, including all parts and accessories, shall be fabricated and finished in a workmanlike manner to produce units free from all defects. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

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## 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of the test set shall be as follows:

a. Preproduction tests

b. Acceptance tests

* 4.3 Preproduction testing.

* 4.3.1 Preproduction test samples. The preproduction test samples shall consist of models representative of the production items. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under supervision of the procuring activity.

* 4.3.1.1 Preproduction test report. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

* 4.3.2 Preproduction tests. Preproduction tests shall consist of all tests described herein under 4.6.

* 4.4 Acceptance tests. Acceptance tests shall consist of the individual tests.

* 4.4.1 Individual tests. Each test set shall be subjected to the following tests as described under 4.6 of this specification:

a. Examination of product

b. Conformance to drawings

c. Functional test

* 4.4.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

* 4.5 Test conditions.

* 4.5.1 Atmospheric conditions. Unless otherwise specified in specific tests, all tests required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25°C (77°F)

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±1°C (2°F). Relative humidity shall be 80 percent or less. Allowance shall be made for deviations from the above conditions if such deviations exist.

* 4.5.2 Test power. Unless otherwise specified in specific tests, the testing power applied shall be 28 volts (V) ±1V direct current (DC). Where alternating current is specified, voltage shall be maintained at 28V ±1V, 400 cycles per second (cps) ±2 percent.

* 4.6 Test methods.

* 4.6.1 Examination of product. Each test set shall be examined to determine compliance with the requirements of this specification, including materials, workmanship, and ease of operation of controls.

* 4.6.2 Conformance to drawings. Each test set shall be inspected to determine strict conformance to engineering drawings and subsidiary drawings.

* 4.6.3 Functional test. Each test set shall be subjected to the functional test in the sequence shown in Table I. An acceptance tester with circuitry in accordance with Figure 1 shall be used.

* 4.6.3.1 Rejection and retest. In the event a test set fails to meet the "indication or response" requirement of any step of test, the test set shall be rejected, and after correction of cause of failure, shall be resubmitted for complete functional testing.

* 4.6.4 Environmental tests. All environmental tests, except vibration and shock tests, shall be conducted with the test set in operational configuration. Vibration and shock tests shall be conducted with the test set in transport and storage configuration. Pretests and operational tests, where required, shall be as specified in 4.6.3.

* 4.6.4.1 High temperature. The test set shall satisfy requirements of the test described in procedure I, method 501.1 of MIL-STD-810, except that the test set shall be stabilized at ambient temperature prior to operation.

* 4.6.4.2 Low temperature. The test set shall satisfy requirements of the test described in method 507.1 of MIL-STD-810.

* 4.6.4.3 Humidity. The test set shall satisfy requirements of the test described in method 507.1 of MIL-STD-810.

* 4.6.4.4 Rain. The test set shall meet requirements of the test described in procedure II, method 506.1 of MIL-STD-810.

* 4.6.4.4.1 Evidence of water penetration shall not be cause for rejection; however, failure to meet the requirements of the functional test of this specification at completion of the rain test shall be cause for rejection.

* 4.6.4.5 Dust. The test set shall meet requirements of the test described in procedure I, method 510.1 of MIL-STD-810.

* 4.6.4.6 Salt fog. The test set shall meet requirements of the test described in method 509.1 of MIL-STD-810.



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TABLE I

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| PREPARATION FOR ELECTRICAL FUNCTION TEST | | | | | | |
| CB-1 | OFF | CB-1 | OFF | Power light | OFF | |
| DB-2 | OFF | CB-2 | OFF | Meter | Reads Zero | |
| CB-3 | OFF | CB-3 | OFF | Signal Indicator | OFF | |
| Power Switch | OFF | DC Power No. 1 | OFF | Fault light | OFF | |
| Meter Switch | No. 2 | DC Power No. 2 | OFF | Continuity Window | S | |
| K2-Relay Signal Switch | OFF | AC Power | OFF | Insulation Resistance Window | S | |
| K1-Relay Switch | OFF | Ground Selector | OFF | | | |
| Circuit Selector | No. 1 | Function Selector | OFF | | | |
| Function Switch | IR | | | | | |
| Polarity Switch | -28V | | | | | |
| Start-Override/Auto Scan Switch | OFF | | | | | |



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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| | | TEST PROCEDURES | | | | |
| 1. Connect Acceptance Tester to AN/GWM-4 Analyzer | | | | | | |
| 2. Connect 28V DC, AC and Ground to Acceptance Tester | | | | | | |
| 3. | | CB-1, CB-2, CB-3 | ON | NOTE: Shall remain energized | | |
| 4. CB-1 | ON | DC Power No. 2 | ON No. 2 | | | No Indication |
| 5. CASE | Momentarily Gnd | | | Meter | Reads 115 ±5 | While Case Grounded |
| 6. | | Ground Selector | 1 | Meter | 115 ±5 | |
| 7.<br>a. CB-2<br>b. Power Switch<br>c. CB-2<br>d. CB-2 | <br>ON<br>ON<br>OFF<br>ON | | | <br>a. Power Light<br>b. Relay in GWM-4<br>c. Power Light<br>d. Power Light | <br>ON<br>Energizes<br>OFF<br>ON | <br><br>Audibly<br><br> |





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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 8. Power light | Press-to-Test | | | Power light | OFF | During Press-to-Test |
| 9. Signal Indicator | Press-to-Test | | | Signal Indicator | ON | During Press-to-Test |
| 10. Fault light | Press-to-Test | | | Fault light | ON | During Press-to-Test |
| 11. Start-Override/Auto Scan Switch | Hold in Scan | | | a. GWM-4 Motor<br>b. Insulation Resistance Window | Is audible<br>"S" | S remains in window |
| 12. Start-Override/Auto Scan Switch<br>CAUTION: Never hold in "Override" Position. | Override S | | | a. Fault light<br>NOTE: This indication occurs periodically during test but has no significance.<br>b. Insulation Resistance Window | lights<br>No. 1 | Momentarily |
| 13. Start-Override/Auto Scan Switch | SCAN 1 to 10 | | | Insulation Resistance Window | Scans to 10 | GWM-4 Motor becomes inaudible after indication 10. |



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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 14. a. K-1 Relay Switch | Depress Momentarily | | | a. Motor | Becomes Audible | While Switches Depressed |
| SIMULTANEOUSLY | | | | | | |
| b. Start-Override/Auto Scan Switch | Scan Momentarily | | | b. Fault Light | ON | |
| | | | | NOTE: Meter reads approximately 165 whenever K-1 is depressed and Meter Switch is in position 2. | | |
| 15. Start-Override/Auto Scan Switch | Override 10 | | | Insulation Resistance Window | No. 11 | |
| 16. Start-Override/Auto Scan Switch | SCAN 11 to 8 | | | a. Insulation Resistance window<br>b. Motor NOTE: During this and subsequent steps motor is audible when indication in IR or Continuity Window is S. | Scans to "Start" | |
| 17. a. Polarity Switch<br>b. Start-Override/Auto Scan Switch | ±28V<br>SCAN | | | Motor | Audible | |
| 18. Start-Override/Auto Scan Switch | Override S | | | Insulation Resistance Window | No. 1 | |





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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 19. Start-Override/Auto Scan Switch | SCAN 1 to 10 | | | Insulation Resistance Scan to 10 Window<br>NOTE: During this and subsequent steps motor becomes inaudible when indication in IR window is "10." | | |
| 20. Start-Override/Auto Scan Switch | Override 10 | | | IR Window | No. 11 | |
| 21. Start-Override/Auto Scan Switch | SCAN 11 to S | | | IR Window | Scan to Start | |
| 22.<br>b. Start-Override/Auto Scan Switch | Override S to S | a. Functional Selector Switch | IR | a. IR Window<br>b.1. Fault Light<br>2. Fault Light | 1 to Start<br>ON<br>OFF | <br>1-7, 13-23<br>8-12, and S |
| 23.<br>a. Polarity Switch<br>b. Repeat Step 22b. | -28V | | | Same as response in Step 22. | | |
| 24.<br>a. Function Switch<br>b. Start-Override/Auto Scan Switch | Continuity<br>SCAN | Ground Selector Function Selector | No. 4<br>"No continuity" | a. Motor<br>b. Continuity Window | Audible<br>S | <br>S remains |



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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM 4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 25. Start-Override/Auto Scan Switch | Override S to 17 | | | a. Continuity Window<br>b. Fault Light<br>c. Fault Light | 1 to 17<br>ON<br>OFF | <br>1, 3-6, 14 and 15<br>2, 7-13, 16 and 17 |
| 26. Start-Override/Auto Scan Switch | Override 17 | | | a. Continuity Window<br>b. Fault Light | 18<br>ON | |
| 27. CB-3 | ON | | | Fault Light | OFF | |
| 28. Start-Override/Auto Scan Switch | Override 18 to 19 | | | a. Continuity Window<br>b. Fault Light | 19-20<br>ON | <br>19 and 20 |
| 29. | | AC Power | ON, then back to OFF | Fault Light | OFF | When AC Power is ON |
| 30. Start-Override/Auto Scan Switch | Override 20 to S | | | a. Continuity Window<br>b. Fault Light<br>Fault Light | 21 to S<br>ON<br>OFF | <br>21 to 23 at S |
| 31.<br>c. Start-Override/Auto Scan Switch | <br>SCAN | a. Ground Selector<br>b. Function Selector | No. 1<br>Continuity | <br>c.(1)Motor<br>(2)Fault Light | <br>Audible<br>OFF | |





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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 32. Start-Override/Auto Scan Switch | Override S | | | Fault Light | ON | |
| 33.<br>b. K-1 Relay | Momentarily Depress | a. Ground Selector | No. 2 | a. Fault Light<br>b.(1) Fault Light<br>(2) Meter | ON<br>OFF<br>Reads between 150 and 200 | While K-1 Switch Depressed |
| 34. Start-Override/Auto Scan Switch | Override 1 | | | a. Continuity Window<br>b. Fault Light | No. 2<br>OFF | |
| 35. Start-Override/Auto Scan Switch | Scan 2 to 7 | | | a. Continuity Window<br>b. Fault Light<br>c. Fault Light | 2 to 7<br>OFF<br>ON | 2 to 6<br>At 7 |
| 36. | | Ground Selector | 3 | Fault Light | OFF | |
| 37. Start-Override/Auto Scan Switch | Scan 7 to 13 | | | a. Continuity Window<br>b. Fault Light<br>c. Fault Light | 7 to 13<br>OFF<br>ON | 7 to 12<br>At 13 |
| 38. | | Ground Selector | 4 | Fault Light | OFF | |



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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 39. Start-Override/Auto Scan Switch | Override 13 | | | a. Continuity Window<br>b. Fault Light | 14<br>ON | |
| 40. K-1 Relay Switch | Momentarily Depress | | | a. Fault Light<br><br><br><br>b. Meter | OFF<br><br><br><br>Reads between 150 and 200 | While relay Switch in Depressed position. |
| 41. Start-Override/Auto Scan Switch | Override 14. Scan to S | | | a. Continuity Window<br>b. Fault Light<br>c. Fault Light | 14 to S<br>OFF<br>ON | 15 to S<br>At 14 |
| 42. Circuit Selector | No. 2 | | | Meter<br>Power Light | Reads Zero<br>OFF | |
| 43. | | a. DC Power No. 1<br>b. DC Power No. 2 | ON<br>ON #1 | Meter<br>Power Light | Reads 115 ±5<br>ON | |
| 44. Function Switch | IR | Function Selector | OFF | | | |





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| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 45. Start-Override/Auto Scan Switch | Override<br>S | | | a. IR Window<br>b. Fault Light | No. 1<br>OFF | |
| 46. Start-Override/Auto Scan Switch | SCAN<br>1 to 10 | | | a. IR Window<br>b. Fault Light | Scan 1 to 10<br>Remain OFF | |
| 47. Start-Override/Auto Scan Switch | Override<br>10 | | | a. IR Window<br>b. Fault Light | No. 11<br>Remain Off | |
| 48. Start-Override/Auto Scan Switch | SCAN<br>11 to S | | | a. IR Window<br>b. Fault Light | Scan 11 to S<br>OFF | |
| 49.<br>a. Polarity Switch<br>b. Repeat steps 45, 46, | ±28V<br>47, and 48 | | | Indications same as steps 45, 46, 47, and 48 | | |
| 50.<br>b. Start-Override/Auto Scan Switch | Override<br>S to S | a. Function Selector Switch | IR | b.(1) IR Window<br>(2) Fault Light<br>(3) Fault Light | S to S<br>ON<br>OFF | 1 to 23<br>At S |
| 51.<br>a. Polarity Switch<br>b. Start-Override/Auto Scan Switch | -28V<br>Override<br>S to S | | | b.(1) IR Window<br>(2) Fault Light<br>(3) Fault Light | S to S<br>ON<br>OFF | 1 to 23<br>At S |



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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 52.<br>a. Function Switch<br><br>b. Start-Override/Auto Scan Switch | Continuity<br><br>Override<br>S to S | Function Selector Switch | No Cont | <br><br>b.(1) Cont Window<br>(2) Fault Light<br>(3) Fault Light | <br><br>S to S<br>ON<br>OFF | <br><br>1 to 23<br>At S |
| 53.<br><br>b. Start-Override/Auto Scan Switch | <br><br>Override<br>S | a. Function Selector | Continuity | <br><br>Cont Window<br>Fault Light | <br><br>No. 1<br>OFF | |
| 54. Start-Override/Auto Scan Switch | Scan<br>1 to S | | | Cont Window<br>Fault Light | S to S<br>OFF | |
| 55.<br>a. Circuit Selector<br>b. Function Switch<br><br>c. Start-Override/Auto Scan Switch | No. 3<br>IR<br><br>Override | Function Selector | OFF | <br><br><br>IR Window<br>Fault Light | <br><br><br>1<br>OFF | |
| 56. Start-Override/Auto Scan Switch | SCAN<br>1 to 10 | | | IR Window<br>Fault Light | Scan to 10<br>OFF | |
| 57. Start-Override/Auto Scan Switch | Override<br>10 | | | IR Window<br>Fault Light | 11<br>OFF | |





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| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 58. Start-Override/Auto Scan Switch | SCAN<br>11 to S | | | IR Window<br>Fault Light | 11 to S<br>OFF | |
| 59. Polarity Switch | ±28V | | | | | |
| 60. Repeat steps 55c, 56, 57, and 58. | | | | Indications same as 55a, 56, 57, and 58. | | |
| 61.<br>b. Start-Override/ Auto Scan Switch | Override<br>S to S | a. Function Selector | IR | IR Window<br>Fault Light<br>Fault Light | S to S<br>ON<br>OFF. | 1 to 23<br>At S |
| 62.<br>a. Polarity Switch<br>b. Repeat step 61b | -28V | | | Indication same as 61b. | | |
| 63.<br>a. Function Switch<br>b. Start-Override/ Auto Scan Switch | Continuity<br>Override<br>S to S | Function Selector | No Cont | Cont Window<br>Fault Light<br>Fault Light | S to S<br>ON<br>OFF | 1 to 23<br>At S |
| 64. | | a. Function Selector | Cont | | | |



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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 64. (Cont'd)<br>b. Start-Override/Auto Scan Switch | Scan 1 to S. Override S, 10 and 11 | | | b.(1) Cont Window<br>(2) Fault Light<br>(3) Fault Light | S to S<br>OFF<br>ON | 1 to 9, 12<br>10 and 11 |
| 65.<br>a. Circuit Selector<br>b. Function Switch | No. 4<br>IR | Function Selector | OFF | | | |
| 66. Start-Override/Auto Scan Switch | Override S and 10. Scan all others | | | IR Window<br>Fault Light | S to S<br>OFF | |
| 67.<br>a. Polarity Switch<br>b. Repeat Step 66 | ±28V | | | Same as 66. | | |
| 68.<br>b. Start-Override/Auto Scan Switch | Override S to S | a. Function Selector | IR | IR Window<br>Fault Light<br>Fault Light | S to S<br>ON<br>OFF | 1 to 8, 10 & 11, 13 to 23<br>9, 12 & S |





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| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 69.<br>a. Polarity Switch<br>b. Repeat step 68b | -28V | | | Same as 68b. | | |
| 70.<br>a. Function Switch<br><br>b. Start-Override/Auto Scan Switch | Continuity<br><br>Override<br>S to S | Function Selector | No Cont | <br><br>Cont Window<br>Fault Light<br>Fault Light | <br><br>S to S<br>ON<br>OFF | <br><br><br>1-8, 10-23<br>9 and S |
| 71.<br><br>b. Start-override/Auto | <br><br>Override<br>S & 7 to 11.<br>Scan all others | a. Function Selector | Cont | <br><br>Cont Window<br>Fault Light<br>Fault Light | <br><br>S to S<br>OFF<br>ON | <br><br><br>1-6,9,12-S<br>7,8, 10&11 |
| 72.<br>a. Circuit Selector<br>b. Function Switch<br><br>c. Start-Override/Auto Scan Switch | No. 5<br>IR<br><br>Override<br>S & 10.<br>Scan all others | Function Selector | OFF | <br><br><br>IR Window<br>Fault Light | <br><br><br>S to S<br>OFF | |



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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 73.<br>a. Polarity Switch<br>b. Repeat step 72c | ±28V | | | Same as 72c | | |
| 74.<br>b. Start-Override/Auto Scan Switch | Override<br>S to S | a. Function Selector | IR | IR Window<br>Fault Light<br>Fault Light | S to S<br>ON<br>OFF | 1-8,10-18, 20-23<br>9, 19, S |
| 75.<br>a. Polarity Switch<br>b. Repeat step 74b | -28V | | | Same as 74b | | |
| 76.<br>a. Function Switch<br>b. Start-Override/Auto Scan Switch | Continuity<br>Override<br>S to S | Function Selector | No Cont | Cont Window<br>Fault Light<br>Fault Light | S to S<br>ON<br>OFF | 1-8, 13-19, 21-23<br>9-12, 20&S |
| 77.<br>b. Start-Override/Auto Scan Switch | Override<br>S, 7, 8<br>Scan all others | a. Function Selector | Cont | Cont Window<br>Fault Light<br>Fault Light | S to S<br>OFF<br>ON | 1-6, 9toS<br>7 and 8 |





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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 78.<br>a. Circuit Selector<br>b. Function Switch<br><br>c. K-2 Relay Signal Switch<br>d. Start-Override/Auto Scan Switch | <br>No. 6<br>IR<br><br>K-2<br><br>Override S and 10 Scan to 22 | Function Selector | OFF | <br><br>IR Window<br><br>Fault Light<br><br>Fault Light | <br><br>S to 22<br><br>ON<br><br>OFF | <br><br><br>22<br><br>S to 21 |
| 79. K-2 Relay Signal Switch | OFF to Signal to K-2 | | | a. Fault Light<br><br>b. Fault Light | OFF<br><br>ON | When K-2 Signal Switch is OFF<br>When K-2 Signal Switch is on Signal or K-2 |
| 80. Start-Override/Auto Scan Switch | Override 22 & 23 | | | Fault Light | OFF | 23 and S |
| 81.<br>a. Polarity Switch<br>b. Repeat steps 78d, 79 and 80 | ±28V | | | Same as steps 78d, 79, and 80, except that Fault Light is ON at 23 instead of 22. | | |



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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 82.<br>a. K-2 Relay Signal Switch<br>b. Start-Override/Auto Scan Switch | OFF<br><br>Override<br>S to S | Function Selector | IR | <br><br>IR Window<br>Fault Light<br>Fault Light | <br><br>S to S<br>ON<br>OFF | <br><br>1-8, 10-23<br>9 and S |
| 83.<br>a. Polarity Switch<br>b. Repeat step 82b | -28V | | | Same as 82b | | |
| 84.<br>a. Function Switch<br>b. Start-Override/Auto Scan Switch | Continuity<br><br>Override<br>S to S | Function Selector | No Cont | <br><br>Cont Window<br>Fault Light<br>Fault Light | <br><br>S to S<br>ON<br>OFF | <br><br>1-8, 10-23<br>9 and S |
| 85.<br>b. Start-Override/Auto Scan Switch | Override<br>S, 7 & 8<br>Scan all others | a. Function Selector | Cont | Cont Window<br>Fault Light<br>Fault Light | S to S<br>OFF<br>ON | 1-6, 9-8<br>7 and 8 |
| 86.<br>b. Power Light | Press-to-Test | a. Ground Selector | No. 5 | Power Light | ON | During Press-to-Test |





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TABLE I (Cont'd)

| OPERATION | | | | RESPONSE | | |
|---|---|---|---|---|---|---|
| GWM-4 | | ACCEPTANCE TESTER | | GWM-4 | | |
| CONTROL | POSITION | CONTROL | POSITION | INDICATOR | INDICATION | REMARKS |
| 87. Meter Switch | No. 1 | | | Meter | Reads Zero | When meter on Position No. 1 |
| 88. Start-Override/Auto Scan Switch | Override S to 9 | | | Cont Window Fault Light | S to 10 | Disregard |
| 89. K-1 Relay | Depress | | | a. Signal Indicator<br>b. Meter | ON<br>Reads Approximately 60 | While K-1 Relay Depressed |
| 90. Start-Override/Auto Scan Switch | Override 10 | | | Cont Window | 11 | |
| 91.<br>a. Repeat step 89<br>b. Start-Override/Auto Scan Switch | Override 11 to S | | | Indication same as 89<br>Cont Window | 11 to S | Disregard Other Indications |
| 92. Power Switch | OFF | DC Power No. 1 | OFF | Power Light | OFF | |
| 93. Meter Switch | 3 | DC Power No. 1 | Leakage | Meter | Initially reads approximately 100 then decays gradually to zero. | |
| 94.<br>a. Reset to "PREPARATION" SETTINGS<br>b. Disconnect Acceptance Tester from AN/CWM-4 | | | | | | |



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4.6.4.7 Altitude. The test set shall meet requirements of method 500.1 of MIL-STD-810.

4.6.4.8 Vibration. The test set shall meet requirements of the test described in procedure XI, method 514.2 of MIL-STD-810.

4.6.4.9 Shock. The test set shall meet requirements of test described in procedure II, method 516.2 of MIL-STD-810.

4.6.4.10 Explosive atmosphere. The test set shall meet requirements of the test described in procedure I, method 511.1 of MIL-STD-810.

4.6.4.11 Fungus resistance. The test set shall meet requirements of the test described in procedure I, method 508.1 of MIL-STD-810. The test set selected for this test shall not have been subjected to the dust test or the salt fog test.

4.6.4.11.1 Evidence of deterioration, corrosion, or support of fungus growth upon completion of this test shall be cause for rejection.

4.6.5 Rejection. Failure of sample test set to satisfy the functional tests during or after environmental test as required in the specific environmental test, shall be considered as failure to satisfy the environmental test and cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packing, and packaging. Preservation, packing, and packaging for shipment shall be in accordance with MIL-E-17555.

5.2 Marking for shipment and storage. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be:

TEST SET, ARMAMENT WIRING SYSTEM, SPECIAL WEAPON AN/GWM-4

6. NOTES

6.1 Intended use. The Test Set, Armament Wiring System, Special Weapon AN/GWM-4 is intended to check aircraft special weapons wiring systems on the F-4, F-100, F-104 and B-52 aircraft.

6.2 Ordering Data. Invitations for bids, contracts, and purchase orders should state the conditions for the following:

6.2.1 Preproduction samples. It is expected that the contract or purchase order will specify that three units representative of the production test set will be submitted to the procuring activity for test and evaluation. The invitation for bids and the contract or purchase order should state the point of inspection for these tests.

6.3 Sources. When requesting publications, refer to both title and number.



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6.4 Definitions.

a. Preproduction tests. Preproduction tests are those tests made on samples which have been submitted by a contractor on a production contract or purchase order to prove, prior to his starting production, that the test set meets the requirements of this specification. Any production prior to the approval of the preproduction samples shall be at the contractor's risk. (See 4.3)

b. Acceptance tests. Acceptance tests are those tests accomplished on equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification and to determine that the units are equivalent in performance and construction to the approved preproduction samples. (See 4.4)

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 79

Review Activity:
Air Force - 15

Preparing Activity:
Air Force - 79

Project No. 1190-F011



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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